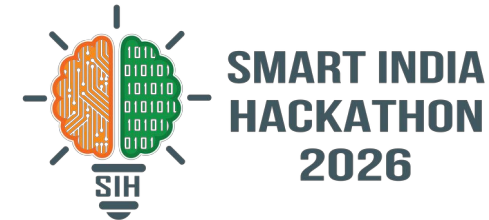
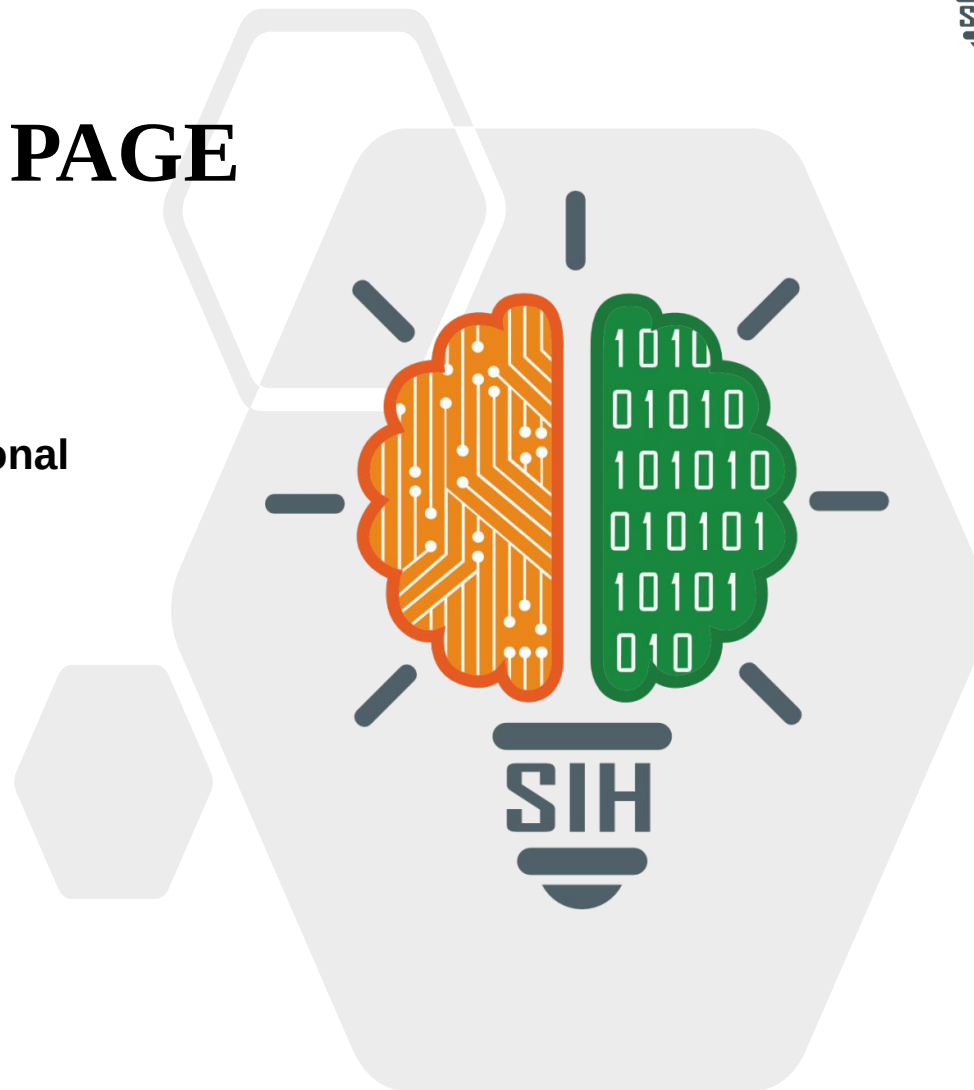


SMART INDIA HACKATHON 2026



TITLE PAGE

- Problem Statement ID – SIH26068
- Problem Statement Title- WeatherGPT: Conversational AI for Weather Forecasting, Alerts, and Climate Information
- Theme- Disaster Management
- PS Category- Software
- Team ID- 118709
- Team Name- Void Pointers



WeatherGPT: ask the sky, trust the answer.

IMD's forecasts in every Indian language, answered from servers in India.



IMD's forecasts, in your language. Every number comes with a receipt.

How it addresses the problem

Weather data is scattered across portals, bulletins, satellite products & forecast systems

One conversational interface: actionable insight in seconds

Generic AI chatbots invent forecasts, even warnings

Accurate: every value verified against IMD & NWP records

English-only, text-heavy portals shut out rural India

Voice-enabled & multilingual via Sarvam AI: 22 Indian languages

WeatherGPT

ગુજરાતી

Surat, Gujarat

કાલે સવારે સુરતમાં વરસાદ પડશે?

FORECAST RAINFALL

0.4 mm

સુરત • કાલે 09:30–12:30

09:30

evidence covers window

12:30

IMD • Green: no warning

PRAMAAN • GFS 12Z • f+18–21 h

cell 21.25N 72.75E • 23:03 IST

સાંજે?

કોઈ ચેતવણી?

તમારો પ્રશ્ન પૂછો...

Concept screen · Gujarati in, Gujarati out



Proposed Solution: seven modules, one conversation

Mausam Setu

IMD-first real-time data

Satya Gate

zero-hallucination verifier

Pramaan Receipt

proof on every answer

Vaani

Sarvam AI: 22 languages, voice

Chetavani Relay

early warning dissemination

Krishi Salah

location-based advisory

Jalvayu Lens

climate trend & historical analysis

PS SIH26068 · 8 / 8 KEY FEATURES COVERED

Real-time

NL querying

GFS / WRF

Early warning

Location

22 languages

Climate trend

Voice

INNOVATION & UNIQUENESS

Zero invented values: Satya Gate checks every number first.

IMD first: IMD's warnings, nowcasts & Mausamgram, beside global models.

Sovereign by design: open models, Indian servers; no question leaves India.

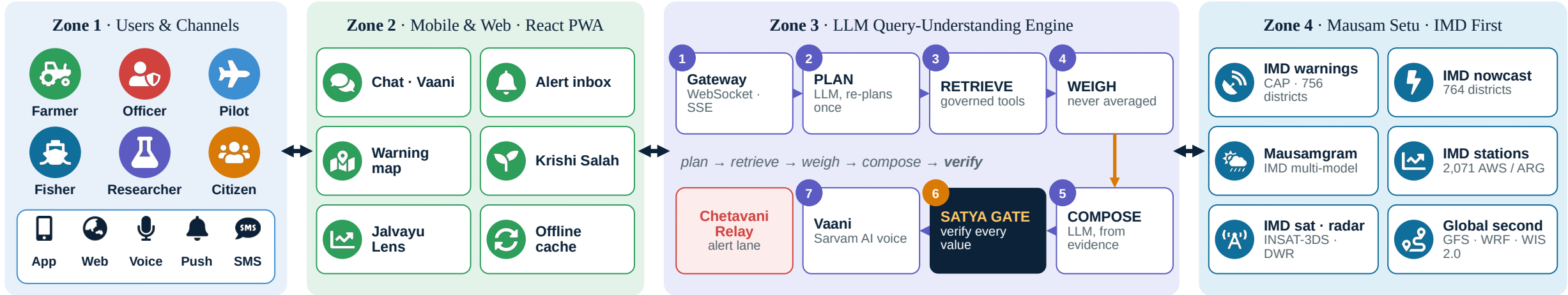
SEE IT LIVE

demo & prototype

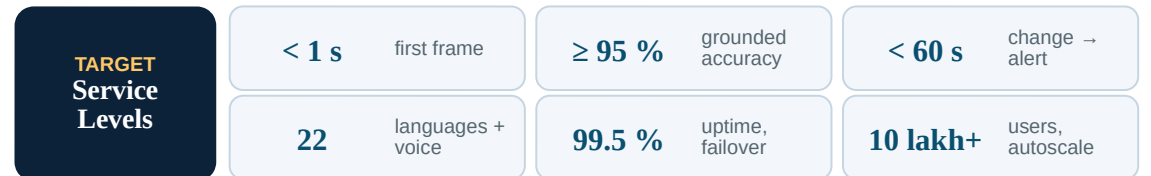
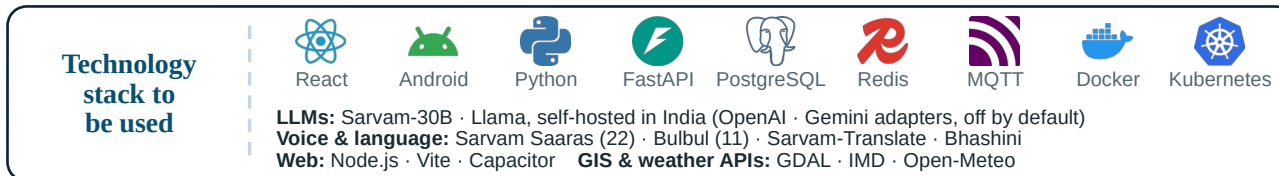
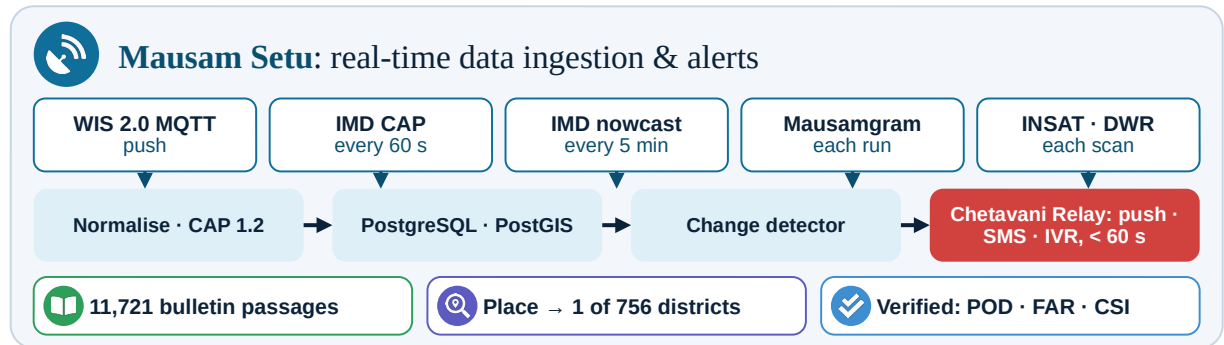
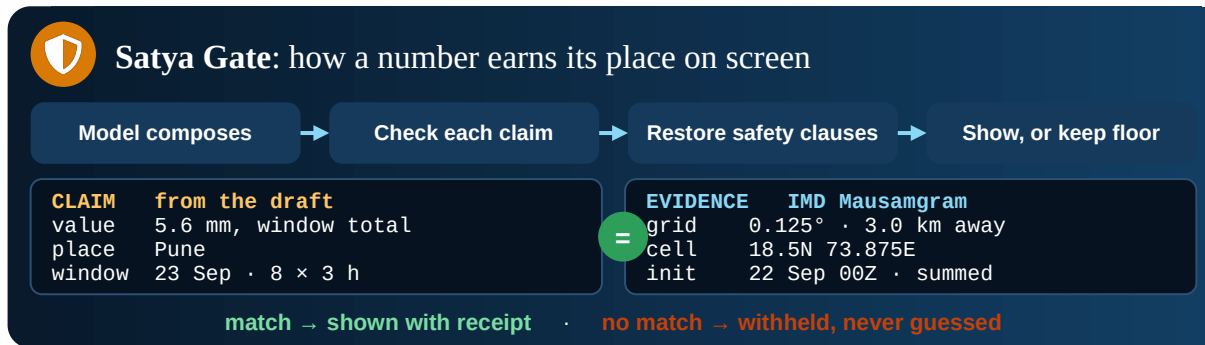


TECHNICAL APPROACH

Technologies to be used · Methodology and process for implementation



THE RULE THAT NEVER BENDS: the LLM plans and speaks; it **never** supplies a number, unit, place, time-window or warning level. Tools own every value.



FEASIBILITY AND VIABILITY

Feasibility analysis · Potential challenges & risks · Strategies to overcome them

Feasibility

90 %

of 306 live scenarios
answered from evidence

- **Proven, available tech:** open-weight Indian LLMs, IMD's own feeds, GFS / WRF, Sarvam speech.
- **Working prototype** today: **1,630 + 596** automated checks, a 21-step gate, **0 failed**.

Viability

100 %

hosted, processed &
stored in India

- **Sovereign & sustainable:** open-weight Indian models (Sarvam-30B) on IndiaAI / NIC compute, IMD-owned.
- Serves **Mission Mausam**; a **chat widget** for IMD's website, apps, UMANG & EOC portals.

Practical Implementation

90 days

to a Gujarat pilot:
every district, gu / hi / en

- **Scalable:** 1 state → every State / UT; **4.3 crore farmers** already get IMD agromet SMS.
- Low-end phones and 2G; day-one owner **IMD Met Centre + State EOC**.

Potential Challenges & Risks

- 01 AI hallucination:** an LLM may invent a number, or even a warning.
- 02 Data access:** IMD's APIs need credentials and approvals.
- 03 Connectivity:** patchy rural networks and feature phones.
- 04 Language fidelity:** translation can distort values and negations.
- 05 Foreign AI dependence:** outages, price shifts, questions sent abroad.

Satya Gate binds every value to a source record; unverified output is never shown.

Public routes to IMD's own data: Mausamgram, 764-district nowcast, 2,071 stations; API adapters switch on with a key.

Offline + SMS / IVR: last-answer cache, tiny PWA payloads, voice call-back.

Value-safe translation: Sarvam-Translate output re-checked for numbers, units & dates, else it stays in source.

Sovereign stack: self-hosted Indian models, a rule-planner floor, zero foreign data egress.

Strategies for Overcoming Challenges

SOVEREIGN Deployment

NATIONAL CORE
IMD / NIC MeghRaj: models, data, audit logs

STATE NODE
State Data Centre + SEOC: languages, alert relay

DISTRICT EDGE
offline node at the DEOC: answers when links fail

CITIZEN DEVICE
PWA cache · SMS · IVR, no data plan needed

STATE-SCALE COST
≈ ₹26 lakh / yr

Biggest line: **2 × H100** inference on **IndiaAI compute** at ₹150 / GPU-hr, 24 × 7. No per-question foreign API fees.

**DATA STAYS
in India**

DPDP Act 2023
district-level location only

CERT-In 2022
logs kept 180 days in India

MeitY cloud
empanelled, India region

Open standards
CAP 1.2 · WIS 2.0 · WFS

NOW
IMD-first answers · installable PWA · streaming

BEFORE FINALE
Live push on a phone · WIS 2.0 lane · WRF

36-HOUR FINALE
SMS / IVR · cyclone-bulletin OCR · IMD API adapters

90-DAY PILOT
Gujarat · all districts · 3 languages

FUTURE SCOPE
WhatsApp bot · panchayat-level forecasts · AI nowcasting · impact-based warnings

IMPACT AND BENEFITS

Potential impact on the target audience · Social, economic & environmental benefits

99 %

of days in Jan–Sep 2025 saw extreme weather in India
Source: CSE 2025 >

4,064

lives lost in those nine months, up 48 % in four years
Source: CSE 2025 >

9.47 M ha

of crops hit, four times the 2022 figure
Source: CSE 2025 >

₹12,500

yearly gain per rain-fed farm household using advisories
Source: NCAER 2020 >

PS OUTCOMES

✓

Faster dissemination of weather information

✓

Improved public accessibility to forecasts

✓

Better disaster preparedness & response

✓

Decision support for agriculture, aviation, marine & urban planning



Farmers: crop-weather advisories at the right hour to spray, sow and irrigate, spoken in their language.



Disaster Managers: flood & cyclone warning dissemination by district: what changed, plus a forwardable brief.



Smart Cities & Urban Planners: smart city weather monitoring: heat, flood risk and air quality.





Aviation & Marine: aviation weather briefing (METAR / TAF); RSMC sea-area & coastal warnings for fishers.



Researchers & IMD: climate analytics for researchers: a century of records, plus what India asks.

Benefits of the Solution



Social

IMD's forecasts reach the **last mile** in 22 languages through **Sarvam AI** voice, SMS & IVR. **Trust** grows when every alert traces to IMD.



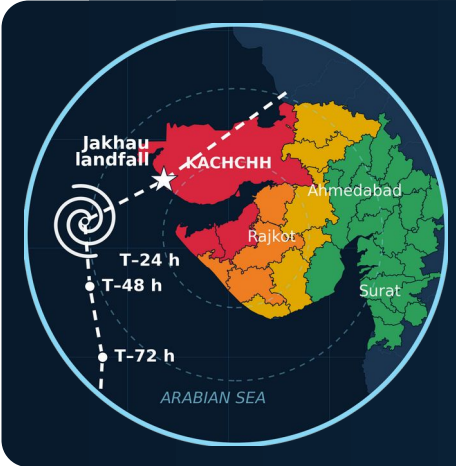
Economic

Advisories are worth **₹12,500 a year** to a rain-fed farm household; run on **Indian government compute**, no foreign fees.



Environmental

Weather-timed spraying & irrigation cut wash-off and water waste; open records power **local adaptation** (SDG 13 · 2 · 11).



DISASTER-DAY STORYBOARD: 72 hours of a cyclone

no warning

watch

alert

warning

Illustrative, modelled on Cyclone Biparjoy (Kutch, June 2023): IMD's warnings moved 1 lakh+ people to safety

T – 72 h



Yellow watch

A farmer asks by voice and hears the rain window plus IMD's own advice: harvest early.

Sarvam voice · Krishi Salah

T – 48 h



IMD raises Orange

The change reaches every subscribed phone in the district within a minute by push, SMS and IVR.

Chetavani Relay

T – 24 h



Red warning

The district officer pulls a what-changed brief with its source and forwards it to every taluka.

Pramaan Receipt

T – 0



Landfall

Networks choke: the cached last answer and the IVR line keep working; fishers hear the RSMC sea-area warning.

Offline cache · IVR

T + 24 h



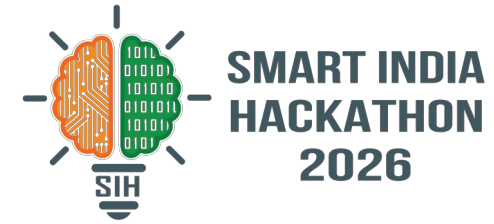
Recovery

Relief teams ask for rainfall totals and river discharge; every figure comes with its source and time.

Mausam Setu · Jalvayu Lens

RESEARCH AND REFERENCES

Details and links of the reference and research work



✓ PS decoded
MoES · IMD needs

✓ 70 sources
live-probed

✓ 5 systems
benchmarked

✓ Architecture
designed

✓ Prototype
306 scenarios run

▶ Now building
live push · WIS 2.0

Existing systems benchmarked

✓ yes · ~ partial · × no

	IMD apps	Weather apps	AI chatbots	WeatherGPT
Data stays in India	✓	×	×	✓
Ask in natural language	×	×	✓	✓
Official IMD warnings	✓	×	×	✓
No invented numbers	✓	✓	×	✓
Source shown on every answer	~	×	×	✓
22 languages + voice	~	~	~	✓
SMS / IVR reach	~	×	×	✓
Forecast · advice · climate in one	×	×	×	✓
IMD + global models, compared	×	×	×	✓

IMD apps studied: [Mausam app](#) · [Meghdoot](#) · [Mausamgram](#)

Official Data, Standards & Research Sources

- **SIH26068**: Ministry of Earth Sciences · India Meteorological Department. sih.gov.in/sih2026PS
- **IMD Mausam**: forecasts, warnings, bulletins. mausam.imd.gov.in
- **IMD GeoServer**: district-warning layers. reactjs.imd.gov.in/geoserver
- **Mission Mausam**: Cabinet note, 2024. pib.gov.in
- **WMO WIS 2.0 · OASIS CAP 1.2**: community.wmo.int
- **NDMA SACHET**: national CAP alert portal. sachet.ndma.gov.in
- **IMD Pune open data**: rainfall & temperature since 1901. dsp.imdpune.gov.in
- **data.gov.in · CERT-In**: open IMD datasets · log directions. data.gov.in

Field & Applied Research

- **70 official sources** live-probed; **public routes to IMD's own products** found (Mausamgram, nowcast, stations).
- **306-scenario live atlas**: 90 % answered from evidence, every refusal classified by cause.
- **8 multi-turn probes**: 7 held end to end.
- Aligned with **Mission Mausam**, the **IndiaAI Mission**, **Bhashini** and the UN **Early Warnings for All**.

WHAT THE RESEARCH CHANGED

Hallucination surveys → **Satya Gate**
RAG literature → **cited bulletin answers**
NCAER advisory value → **Krishni Salah**
CAP 1.2 · WIS 2.0 → **Chetavani Relay**

- **NOAA GFS · Open-Meteo**: numerical weather prediction. open-meteo.com
- **Lewis et al., 2020**: Retrieval-Augmented Generation. [arXiv:2005.11401](https://arxiv.org/abs/2005.11401)
- **Ji et al., 2023**: hallucination in NLG, ACM CSUR. [arXiv:2202.03629](https://arxiv.org/abs/2202.03629)
- **NCAER, 2020**: value of weather advisories. [NCAER2020.pdf](https://ncaer2020.pdf)
- **CSE · Down To Earth**: Climate India 2025. cseindia.org
- **Sarvam AI**: sovereign LLM, IndiaAI Mission; Saaras, Bulbul, Translate. sarvam.ai/models
- **IndiaAI compute**: sovereign GPUs for Indian models. indiaai.gov.in

See it · Run it · Verify it



Scan for the demo & prototype

Source code: github.com/Abhichandani-Yash-Manish/WeatherGPT_

IMD's evaluation parameters → our answer

Accuracy & relevance

Satya Gate · 90 % of 306 live scenarios answered

Response latency

first frame streamed in 0.4 s, measured

Multilingual capability

Sarvam AI: 22 languages in, 11 spoken back

UI & accessibility

voice-first, WCAG 2.1 AA, low-end phones

Scalability & innovation

stateless API on Kubernetes · Pramaan Receipt

Real-time integration

IMD warnings, nowcast, Mausamgram · WIS 2.0

Voice for rural access

Sarvam Saaras + Bulbul voice, IVR call-back

